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$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

$$E[h(X)] = \sum_{\text{all vals of } X} h(x) p(x)$$

For $[h(X)] = X$, $n = 3$, $p = \frac{1}{2}$

$$E[X] = \sum_{\text{all vals of } X} X * \binom{3}{x} 0.5^x (0.5)^{3-x}$$

$$E[X] = \sum_{\text{all vals of } X} X * \binom{3}{x} 0.5^3$$

$$E[X] = \sum_{\text{all vals of } X} X * \binom{3}{x} \frac{1}{8}$$

$$P(0) = 1/8$$

$$P(1) = 3/8$$

$$P(2) = 3/8$$

$$P(3) = 1/8$$

$$E[X] = 12/8$$